Curriculum Vitae

Jason J Wang
Nashville, TN 37203
Email: jason.j.wang@vanderbilt.edu

Education

Medical Scientist Training Program, Vanderbilt University School of Medicine Jun 2024 – Present
Undergraduate student, College of Arts and Sciences, University of Pennsylvania, Aug 2021 – May 2024
Major in Biophysics, minors in Chemistry and Computational Neuroscience
GPA: 3.96 / 4.0
Undergraduate student, Lewis Honors College, University of Kentucky, June 2020 – May 2021
Double major in Chemistry and Biology, minor in Mathematics
GPA: 4.0 / 4.0

Research Experience

Graduate student in the lab of Dr. Carlos Oliver at Vanderbilt University in the Center for AI in Protein Dynamics (CAIPD).

- Combining deep learning models and MD simulation tools to probe the dynamics of different modes of serotonin transport

Rotated for 100 hours in the lab of Dr. Lisa Monteggia at at Vanderbilt University for three weeks in the spring of 2026.

- Learning basic lab techniques including neuron culture, hippocampal dissections, and whole cell patch clamping

Rotated for 150 hours in the lab of Dr. Stephanie Wankowicz at Vanderbilt University for three weeks in the summer of 2025.

- Identifying patterns of water molecule placement on cyclin-dependent kinases to inform design of more targeted inhibitors

Rotated for 200 hours in the lab of Dr. Jens Meiler at Vanderbilt University for four weeks in the summer of 2024.

- Using modified versions AlphaFold2 and editing inputs to produce conformations of protein-protein interactions for bifunctional molecule design

Worked 1100 Hours in the lab of Dr. Baumgart at the University of Pennsylvania starting as a sophomore in Fall 2021.

- Using optical trapping to determine physical properties of vesicles
- Writing and documenting new code for operating the optical trap
    - Controlling external motor and camera to perform and view the experiment
- Micro pipetting and microscopy to view GUVs in chambers and on camera
- Determining reaction kinetics of vesicles using light scattering fluorimetry
- Developing tracking algorithms and simulations for bead images
- Developing particle detection algorithms for fluorescence microscopy

Performed a 400-hour research project as an intern with the University of Tennessee at Chattanooga over the summer of 2021 (NSF-sponsored REU in interdisciplinary computational biology).

- Computing quantitative measures of the topology of proteins in their native state using Python
- Correlating measures with the folding rate using Pandas and SciPy
- Creating plots using Matplotlib/Seaborn to visualize results
- Writing research results and analysis, publishing, and presenting at a conference

Conducted 40 hours of exploratory research within a group at the Math Lab at the University of Kentucky under the mentorship of Dr. Manon in the Department of Mathematics as a freshman.

- Reviewed literature related to drawing voting districts to quantify gerrymandering
- Gathered voting and districting data from online
- Tried to use published python library for computation

Performed a 360-hour Capstone research project as a junior/senior in high school at the University of Kentucky under the mentorship of Dr. Sathaye of the Department of Mathematics.

- Combined methods from two sources and applied them to a specific problem
- Created a computer program to aid with computation
- Wrote a comprehensive capstone research paper to complete project

Research Publications

- Wang, J. and Panagiotou, E., 2022, The protein folding rate and the geometry and topology of the native state, *Scientific Reports* **12** 6384 [Link to full paper]

Presentations

- Invited talk at Joint Mathematics Meetings, Undergraduate Research Activities in Mathematical and Computational Biology Session. April 8, 2022 [Full presentation here]

Relevant Coursework

Courses at Penn (approx. grade), *graduate level:

- Introduction to probability (100/A+), Inferential Statistics (98/A+), Stochastic Processes (95/A)
- Partial Differential Equations (97/A), Physical Models of Living Systems (94/A), Laboratory Electronics (98/A+), Physical Networks* (A), Applied Machine Learning* (93/A+)
- Intro to Neuroscience (97/A+), Neurobiology (A), Cell Biology (91/A-), Theoretical and Computational Neuroscience* (97/A), Brain Computer Interfaces* (A)
- Biological Chemistry (95/A), Physical Chemistry I (96/A), Physical Chemistry II (95/A), Statistical Mechanics* (97/A+), Chemical Biology* (A), Independent Research (A)

Courses at UK (approx. grade):

- Principles of Genetics (99/A)
- Multivariable Calculus (A), Ordinary Differential Equations (100/A), Matrix Algebra (100/A)

Honors and Awards

Barry Goldwater Scholarship, University of Pennsylvania, March 2023
Dean's List, College of Arts & Sciences, University of Pennsylvania, 2021-2024
Dean's List, College of Arts & Sciences, University of Kentucky, Fall 2020 & Spring 2021

Technical skills

Python Programming, about 1200 hours of experience (research internship, courses, personal projects)

- Computing and simulation in NumPy and SciPy, data analysis using Pandas
- Data and image visualization in Matplotlib and Seaborn
- Applied machine learning with sklearn, Tensorflow, and PyTorch

LabVIEW Programming, about 500 hours of experience with the Baumgart research group

- NI Vision for machine vision
- Data input/output and instrument control

LaTeX Typesetting, about 200 hours of experience

- Two research papers, other homework assignments
- Graphics using TikZ

Volunteer/Extracurricular Experience

Senior Analyst, Wharton Analytics Fellows, 2022-2023

- Data analysis for Fox Entertainment, an MLS team, and Penn athletics
- Presenting findings at business meetings
- Data Analytics workshops and speaker events